ICT2152 · Object Oriented Analysis and Design

Worked Examples: Responsibility and Coupling

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

Assign each responsibility to the object with the information needed to perform it. High cohesion keeps related behavior together; low coupling limits dependencies between parts. Dependency inversion lets policy depend on an abstraction.

2. Example and reasoning

If a screen calculates final grades, it couples presentation to academic policy. Moving the calculation to a domain service makes the rule reusable by reports and easier to test.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “worked examples: responsibility and coupling” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.